

SOFTWARE ENGINEERING DATAFLOW DESIGN MODEL

Group-25

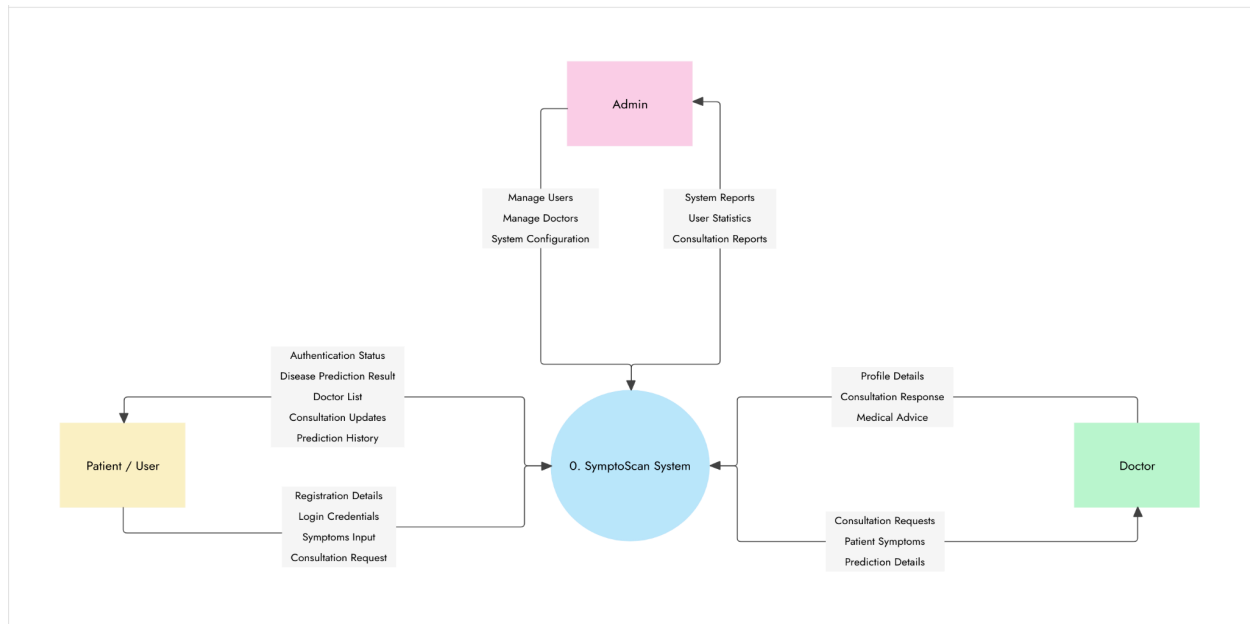
202512012 - Jainam Shah

202512059 - Jay Shah

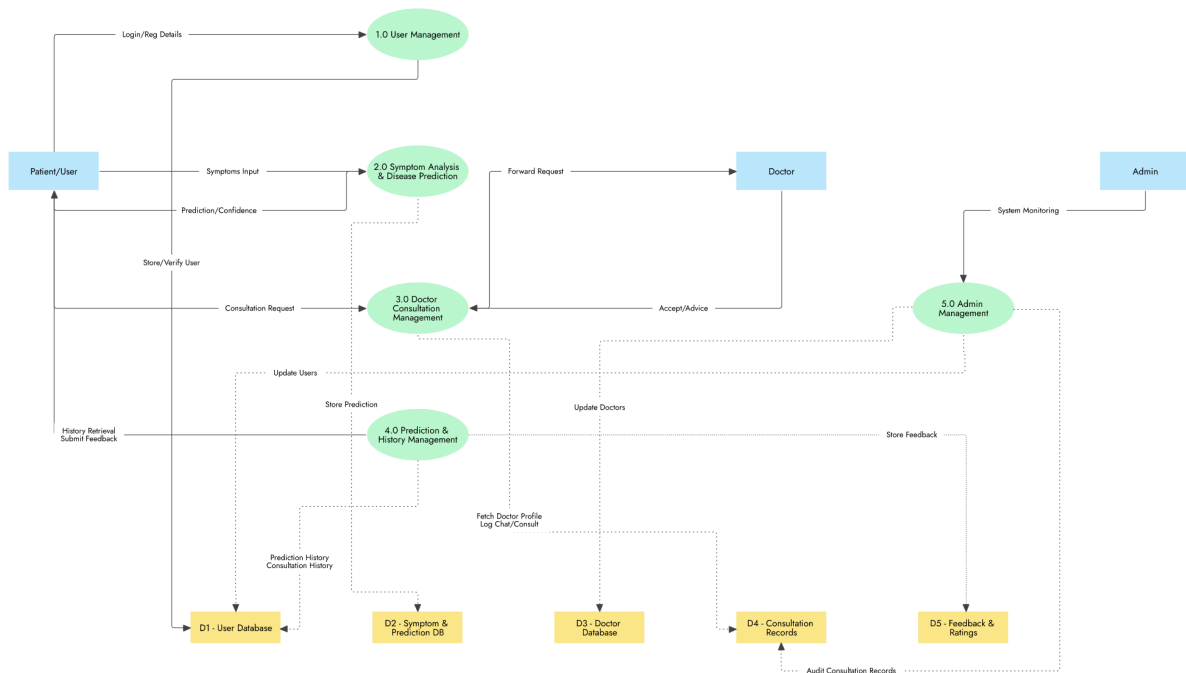
202512100 - Priya J Shah

202512105 - Priya D Shah

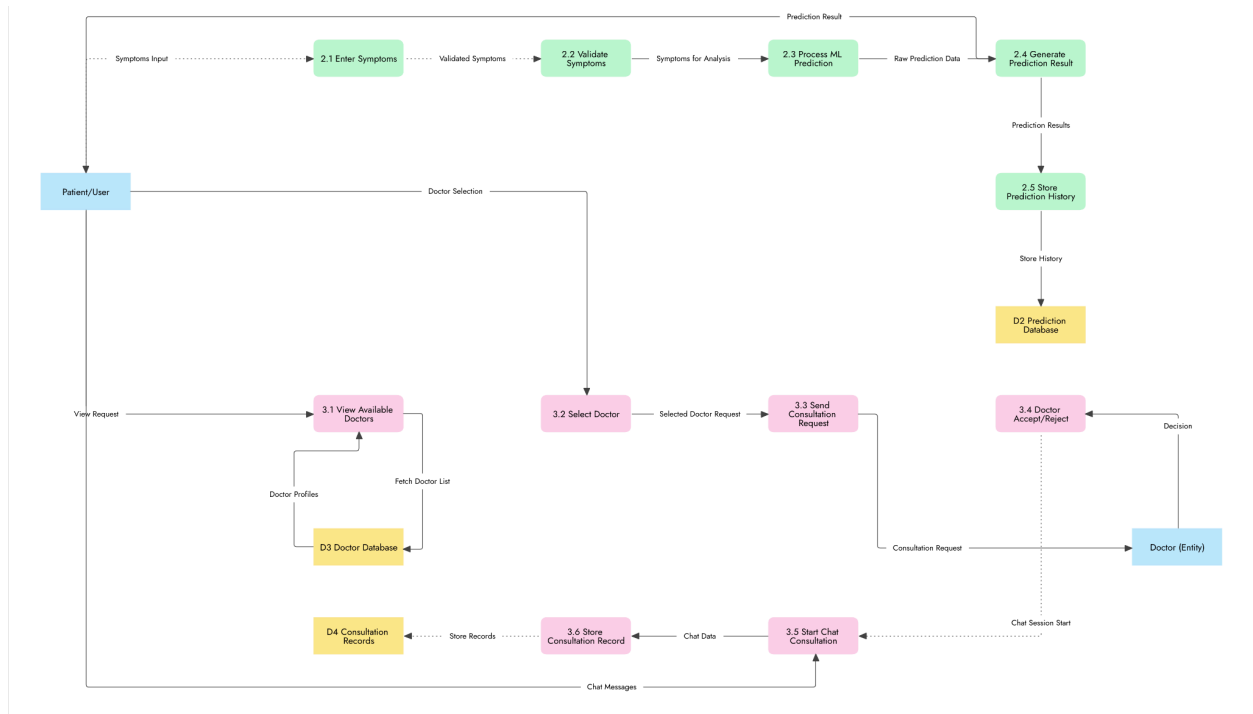
● DFD Level 0



• DFD Level 1



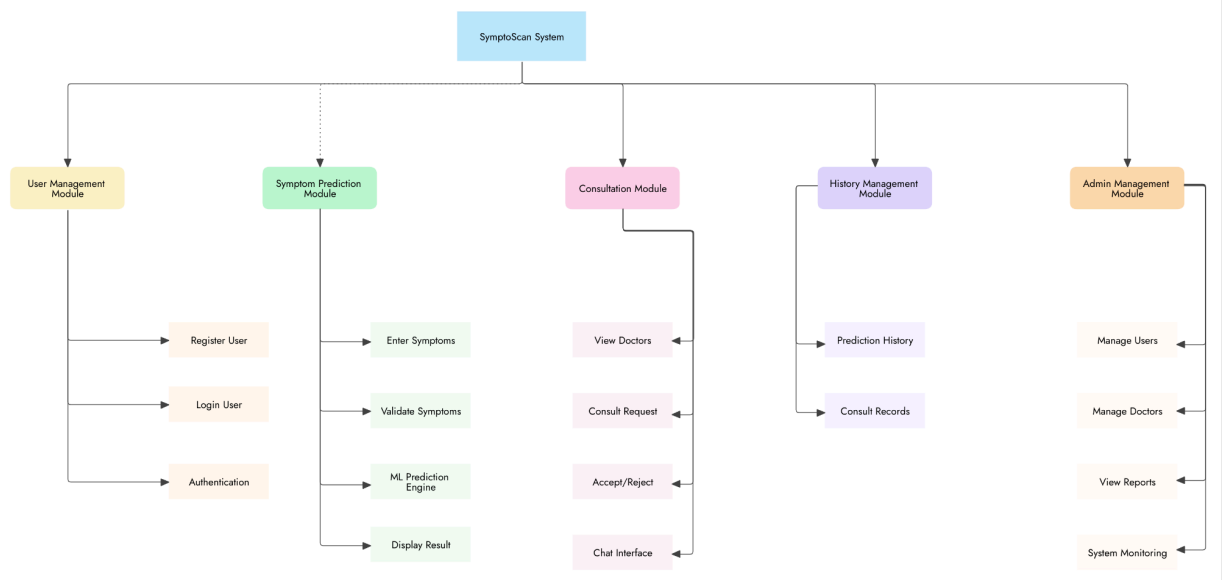
• DFD Level 2



• Data dictionary

Data Element	Description	Data Type	Source / Input	Used By Module
User_ID	Unique identifier assigned to each user in the system	Integer / UUID	System Generated	User Management
Name	Full name of the user or doctor	String	User Input	User Management
Email	Email address used for login and communication	String	User Input	User Management
Password	Encrypted password used for authentication	String (Encrypted)	User Input	Authentication System
Role	Defines whether the user is Patient, Doctor, or Admin	String	System / User Registration	User Management
Symptoms	List of symptoms entered by the patient for disease prediction	Array / List of Strings	Patient Input	Symptom Prediction
Prediction_Result	Disease predicted by the ML model based on symptoms	String	ML Prediction Engine	Prediction System
Confidence_Level	Probability or confidence percentage of predicted disease	Float / Percentage	ML Model	Prediction System
Doctor_ID	Unique identifier assigned to doctors	Integer / UUID	System Generated	Doctor Management
Consultation_Request	Request sent by patient to consult a doctor	Text / Record	Patient Input	Consultation System
Chat_Message	Messages exchanged during doctor-patient consultation	Text	Patient / Doctor	Chat System
Consultation_Record	Stored record of completed consultation session	Structured Record	System Generated	History Management
Feedback	Rating and feedback given by patients for doctors	Text / Integer Rating	Patient Input	Feedback System
Prediction_History	Stored history of all predictions made for a user	Structured Record	System Generated	History Management

● Module structure chart



● Traceability Matrix

Requirement ID	Requirement Description	User Role	System Module	Related Process/Feature
US001	Patient can register using email and password	Patient	User Management	Registration System
US002	Patient can log in securely	Patient	User Management	Authentication System
US003	Doctor can register and manage profile	Doctor	User Management	Doctor Profile Management
US004	Patient can enter symptoms for analysis	Patient	Symptom Prediction	Symptom Input Interface
US005	System predicts disease based on symptoms	System	Symptom Prediction	ML Prediction Engine
US006	Patient can view prediction history	Patient	History Management	Prediction History
US007	Patient can view available doctors	Patient	Consultation Module	Doctor List Interface
US008	Patient can send consultation request	Patient	Consultation Module	Consultation Request System
US009	Doctor can accept or reject consultation requests	Doctor	Consultation Module	Consultation Management
US010	Doctor and patient can chat in real time	Doctor & Patient	Consultation Module	Chat Interface
US011	System stores consultation records	System	History Management	Consultation Record Storage
US012	Admin can manage users and doctors	Admin	Admin Management	User & Doctor Management
US013	Admin can monitor system statistics	Admin	Admin Management	System Reports